PRACTICE: Slope-Intercept Form

y = mx + b

Name: _____ Date: _____ Period: _____

SLOPE: $m = \frac{y2 - y1}{x2 - x1} = \frac{rise}{run}$ Horizontal --> m = 0

SLOPE-INT: y = mx + b (m = slope, b = y-intercept) Vertical --> m = undefined

① Calculate Slope from Two Points

1. Find the slope: (2, 3) and (5, 9) m = _____

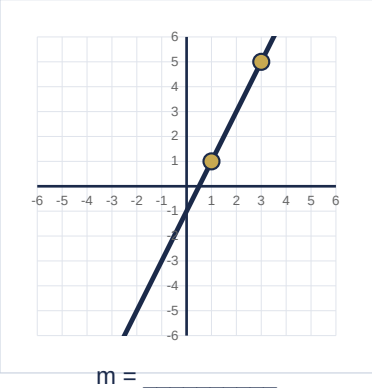
2. Find the slope: (-1, 4) and (3, -2) Write as a simplified fraction! m = _____

3. Find the slope: (4, 1) and (4, 7) m = _____

② Calculate Slope from a Graph

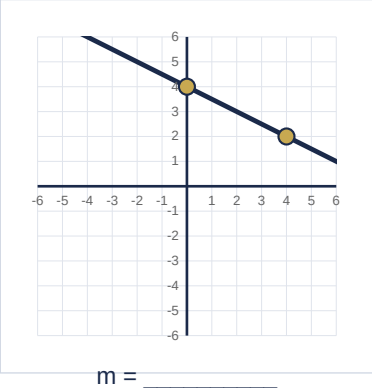
Identify two points on each line. Calculate rise over run.

4.



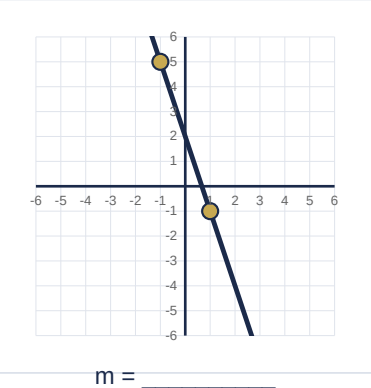
m = _____

5.



m = _____

6.



m = _____

③ Identify Slope & Y-Intercept

7. y = 3x + 5 m = _____ b = _____

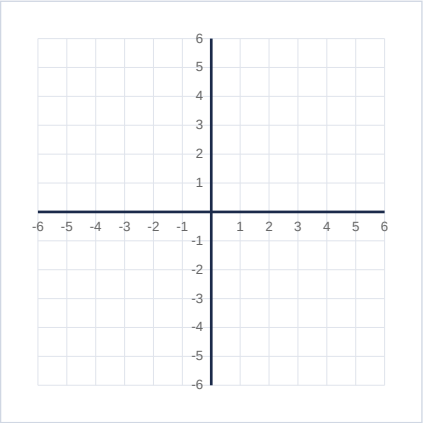
8. $y = -\frac{1}{2}x + 2$ m = _____ b = _____

9. y = -4x m = _____ b = _____

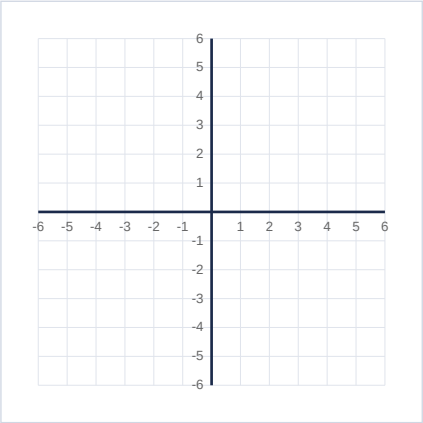
④ Graph Equations in Slope-Intercept Form

Plot the y-intercept. Use slope (rise over run) to find a second point. Draw the line through both.

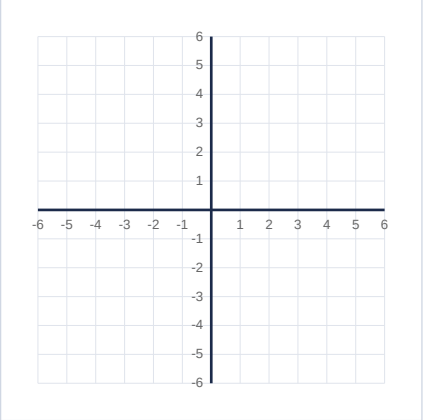
10. $y = 2x - 1$



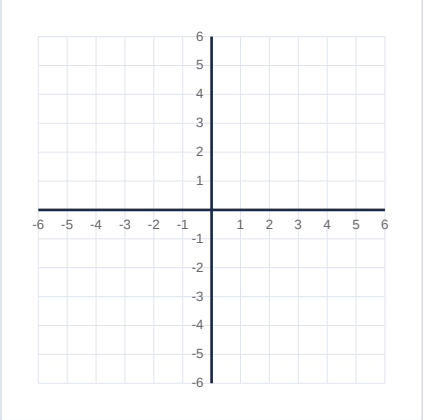
11. $y = -\frac{1}{2}x + 3$



12. $y = -x + 4$



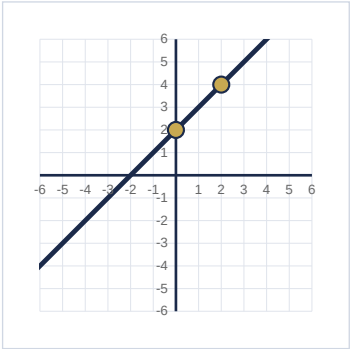
13. $y = \frac{1}{3}x + 1$



⑤ Write Equations from a Graph

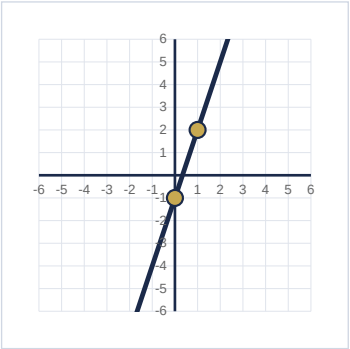
Find the y-intercept and slope. Write the equation in y = mx + b form.

14.



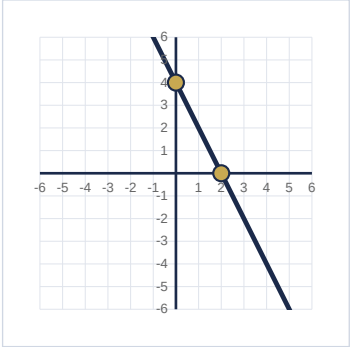
y = _____

15.



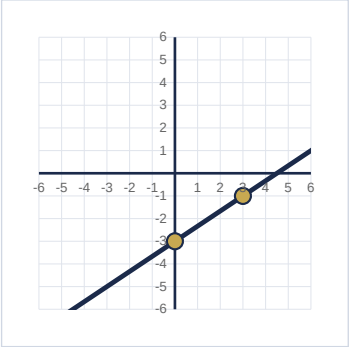
y = _____

16.



y = _____

17.



y = _____